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Coordinated Q-learning based Multi-hop Routing for UAV-assisted communication[☆]

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ABSTRACT

Unmanned Aerial Vehicle (UAV) assisted communication is gaining prominence as a vital solution for establishing effective emergency communication during disaster management operations. UAVs are essential for enhancing and expanding communication systems, acting as relays to boost data transmission to ground stations, extend network coverage, and provide connectivity. However, the dynamic and resource-limited nature of aerial networks necessitates robust routing mechanisms to facilitate seamless data dissemination. While existing Q-learning-based routing protocols are adaptive to changing network conditions and resilient to failures, they often lead to suboptimal network-wide decisions due to UAVs operating independently, each maximizing its gains. This paper proposes a novel Coordinated Q-learning-based Multi-hop Routing (CQMR) algorithm for multi-UAV networks. To the best of our knowledge, this is the first time a routing algorithm introduces UAV coordination for data routing through utility function approximation with a message-passing scheme, enabling the selection of globally optimal joint actions. This novel approach meticulously considers a comprehensive set of parameters for data routing, including minimizing the expected number of hops to the destination, monitoring energy usage, maintaining network connectivity, preventing UAV collisions, and supporting adaptive network reorganization. This integrated consideration of multiple factors positions the proposed solution as superior to existing work, offering a uniquely robust and highly effective strategy for UAV-assisted communication in dynamic, resource-constrained environments, such as emergency scenarios. CQMR builds upon and extends the Improved Q-learning-based Multi-hop Routing (IQMR) algorithm, demonstrating a 12.47% increase in energy efficiency and a 13.34% higher success rate in data transmission compared to IQMR while requiring 40% fewer hops to reach the destination.

1. Introduction

Natural disasters, such as earthquakes, can have devastating impacts, often resulting in the partial or complete destruction of existing terrestrial communication networks. When the cellular networks are disrupted, establishing an alternative wireless infrastructure becomes essential for supporting search and rescue operations. In this context, there is a need for a robust, rapid, and efficient emergency surveillance and communication system to ensure the reliable transfer of disaster information to expedite recovery operations. Unmanned Aerial Vehicles (UAVs), with their high manoeuvrability, adaptable deployment, and versatile

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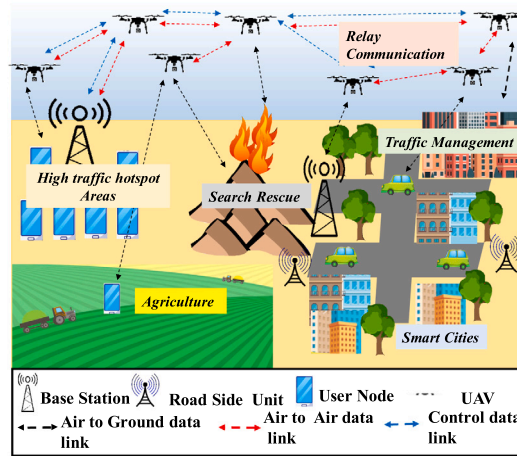


Fig. 1. Illustration of diverse applications supported by the UAVs.

communication capabilities, are well-suited for such tasks [1]. They can effectively respond to emergencies by monitoring disaster areas, searching for lost persons, providing timely rescue assistance, delivering rescue aids to areas otherwise inaccessible by humans, and assessing the extent of damage to people and property [2]. In this paper, disaster management is presented as a motivating use case to demonstrate the relevance and effectiveness of the proposed routing strategy in complex, dynamic environments. However, it is important to note that the core algorithm—based on coordinated Q-learning—is not dependent on the disaster context and is broadly applicable to a variety of UAV-assisted communication scenarios. Beyond post-disaster response, UAVs play an increasingly vital role across a wide range of domains [3], including: (i) precision agriculture, (ii) road traffic management, (iii) on-demand wireless communication provision in high-traffic hotspots, (iv) UAVs as relays to extend coverage, enhance signal strength, and facilitate communication in areas with weak connectivity or remote locations, and (v) smart city applications encompassing surveillance, monitoring, logistics, and urban planning as illustrated in Fig. 1.

To ensure reliable transmission, efficient resource utilization, and seamless communication in dynamic and resource-constrained aerial networks, a robust data routing mechanism is essential. Given the similarities among UAV networks, Mobile Ad Hoc Networks (MANETs), and Vehicular Ad Hoc Networks (VANETs), existing protocols from MANETs and VANETs, along with their variations, have been investigated for potential adaptation in aerial networks [4]. However, these routing protocols typically exhibit reduced efficiency when applied to UAV networks. The routing complexity in UAV networks stems from high mobility, dynamic topology, varying altitudes, sparse deployment, intermittent connectivity, coverage limitations, and resource constraints, which necessitates specialized routing strategies explicitly tailored for UAV networks [3].

In light of these challenges, the focus is on Q-learning-based routing protocol known for its adaptive and predictive abilities that align well with the dynamics of UAV networks [5]. Q-learning enables UAVs to make informed decisions by considering both the current state of the network and their learned experiences, thereby optimizing routing performance across varying operational conditions.

Current Q-learning-based routing protocols predominantly focus on individual parameters such as delay [6], data transmission rate [7], energy constraints [8], and link quality [9]. However, in scenarios with intermittent links and sporadic end-to-end connectivity, it is crucial to adopt a more comprehensive approach that considers multiple objectives. Additionally, existing Q-learning-based routing methods often treat UAVs as autonomous entities, each maximizing its utility through distributed and independent learning, which overlooks the need for a holistic system policy [10]. In multi-UAV networks, the system's behaviour depends on collective actions of all UAVs. Generally, in multi-agent systems, state transitions depend on the current state and the actions of other agents [11]. When a UAV is unaware of the decisions made by other UAVs, future states become unpredictable, leading to sub-optimal routing decisions, reduced reliability, and compromised network stability. Therefore, coordinating UAVs to jointly select a routing policy that optimally benefits the entire network is crucial.

This paper, considers a scenario in which UAVs are to survey disaster-affected areas, gather critical information, and relay the collected data to a Terrestrial Base Station (TBS), as depicted in Fig. 2. To facilitate the seamless transmission of this data, a novel Coordinated Q-learning-based Multi-hop Routing (CQMR) algorithm that addresses the unique challenges of data routing in UAV networks is presented. The proposed algorithm integrates multiple network parameters, including energy consumption, coverage, inter UAV collision avoidance, and the expected number of hops to the destination, while dynamically determining UAV trajectories without pre-defined paths.

The CQMR algorithm facilitates coordination among UAVs, each operating with its own set of observations and potential actions. UAVs use utility function approximation combined with a message-passing scheme to jointly select actions and optimize overall network performance. This technique simplifies managing complex systems like UAV networks by decomposing the overall utility function representing the benefits of different actions into simpler, local utility functions. Each local utility function corresponds

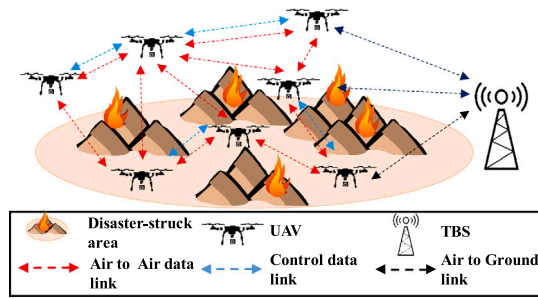


Fig. 2. UAVs surveying disaster areas and relaying data to a Terrestrial Base Station (TBS).

to a small group of UAVs, making the problem more manageable. The message-passing scheme enables UAVs to communicate and coordinate their actions, ensuring collaborative efforts towards a globally optimal outcome despite having only local information. The algorithm then employs a $Q(\lambda)$ -learning framework to select the most efficient next-hop node for data transmission.

To the best of our knowledge, this is the first time a routing algorithm introduces UAV coordination for data routing through utility function approximation with a message-passing scheme, enabling the selection of globally optimal joint actions while systematically incorporating a broad spectrum of parameters for precise routing optimization. The contributions of this paper include:

- (1) A novel Coordinated Q-learning-based Multi-hop Routing (CQMR) algorithm for multi-UAV networks applicable to emergency scenarios like disaster management operations. The algorithm focuses on selecting optimal next-hop UAVs to minimize the expected number of hops to the destination (TBS), while addressing energy management, maintaining network connectivity, avoiding UAV collisions, and supporting adaptive network reorganization.
- (2) A coordination mechanism for UAVs to determine optimal actions within a parametric representation framework, avoiding the need for exhaustive evaluation of every potential action in the joint action space. This strategy uses utility function approximation, where the overall utility function is estimated as a linear combination of local utility functions for small groups of UAVs. Followed by a message-passing scheme implemented as a negotiation procedure during action selection.
- (3) A coordinated $Q(\lambda)$ -learning framework to prioritize the relevant experiences for decision-making. The $Q(\lambda)$ approach learns from interactions with the environment and relevant UAVs, continuously updating policies to identify the most efficient next-hop node for data transmission.
- (4) Comparison of the proposed CQMR algorithm with its uncoordinated counterpart, the Improved Q-learning-based Multi-hop Routing (IQMR) algorithm. CQMR exhibits less disturbance and better responsiveness to abrupt changes in network conditions. CQMR shows a 12.47% increase in energy efficiency, a 13.34% higher data transmission success rate, despite a packet loss rate of 3.33%, while requiring approximately 40% less expected hops to the destination.

The remainder of this paper is organized as follows: Section 2 details the related work on methods designed for UAV data routing, focusing on Q-learning based protocols. Section 3 describes the system model and the propagation channel. Section 4 presents the proposed CQMR algorithm, detailing the four modules of CQMR: (i) coordination set formulation, (ii) state, action, and reward space design, (iii) joint action selection mechanism, and (iv) coordinated $Q(\lambda)$ learning framework. Section 5 analyzes the performance of the proposed CQMR algorithm under varying system parameters and compares it with the IQMR algorithm. Finally, Section 6 concludes the paper and discusses directions for future work.

The parameters and symbols used in this paper are listed and explained in Table 2.

2. Related work

The integration of UAVs in communication networks has attracted considerable attention, particularly for their ability to establish communication, enhance connectivity and manage resources. This section details the existing Q-learning-based routing protocols designed for UAV networks, delves into UAV coordination mechanisms, and concludes with a comparison between the proposed CQMR algorithm and other existing methods.

2.1. Existing Q-learning based protocols

Q-learning has emerged as a powerful tool in optimizing routing protocols for UAV networks, particularly in dynamic and unpredictable environments, such as those encountered during disaster management deployments. The Q-learning protocols adaptively learn from the network environment, enabling UAVs to make informed routing decisions without requiring prior knowledge of network dynamics. This subsection reviews key Q-learning-based routing protocols developed for UAV networks, highlighting their approaches, and limitations in addressing the unique challenges of aerial networks. Q-learning-based Multi-hop

Routing (QMR) [6] attempts to improve the overall routing performance by learning optimal paths through the network considering both delay and energy consumption. However, QMR struggles with scalability as network size increases, leading to high routing overhead and inefficiencies in dynamic environments. Q-FANET [9] represents a significant step forward by combining elements from QMR and Q-Noise+ (Q-Noise+ is a Q-learning-based noise reduction technique to stabilize the spectrum allocation process) [12] to address high-mobility challenges in flying networks. It demonstrates a reduction in delay and jitter while improving packet delivery ratios. However, its scalability and adaptability to varying network conditions remain concerns. Similarly, QTAR [8] (Q-learning-based Topology-Aware Routing) enhances routing by incorporating two-hop neighbour information while considering transmission delay and varying network conditions. While QTAR achieves better routing performance, it also struggles with high routing overhead and scalability issues.

Building on the idea of adapting Q-learning for dynamic networks, Topology-Aware Resilient Routing with Adaptive Q-learning (TARRAQ) [13] integrates adaptive sensing intervals and link duration estimation using the Kalman filter. This approach aims to reduce delay and energy consumption. Nevertheless, TARRAQ faces high routing overhead and limited scalability. Q-learning-based Geographic Routing (QGeo) [10], Q-learning-based Geographic Routing (QLGR) [14], and Traffic-Aware Q-network Geographic Routing scheme based on Greedy Perimeter Stateless Routing (TQNGPSR) [15] incorporate Q-learning into geographic routing. While they enhance routing decisions based on geographic information, they are limited by fixed learning parameters and single mobility models, impacting their adaptability. This challenge is echoed in Q-learning-based Spatial Routing Protocol (QSRP) [16], which utilizes spatial information for routing but suffers from high routing overhead and low scalability. Further, Three-Dimensional Q-learning-based Routing (3DQ) [17] combines greedy and store-carry-forward techniques to address 3D mobility but suffers from high routing overhead. Lastly, the Q-learning-based Cross-Layer Routing Protocol (QLCLRP) [7] introduces a cross-layer approach by integrating carrier sense multiple access with multi-channel synchronization to ensure reliable data transmission in UAV networks. Though this method improves packet arrival ratios and reduces delays, but faces challenges with integrating advanced learning techniques. However, all these existing Q-learning-based routing protocols operate in an autonomous manner without coordination; the independent routing decisions made by each UAV may not contribute to the system's global good. Therefore, coordinated efforts are required among UAVs to avoid sub-optimal results and enhance overall network performance.

2.2. Mechanisms for coordinated UAV operations

A typical approach to addressing the coordination problem is to model the system as a Markov Decision Process, where the action represents a joint action for all UAVs, and the reward is the cumulative reward for all UAVs [19]. A multi-UAV multi-task packet forwarding procedure is formulated as an MDP process to determine the optimal task schedule, maximizing global performance gain for packet forwarding link generation and also ensuring coordination between the UAV fleet, guided by the ground base station [20]. Further, a Q-learning-based Routing protocol Inspired by Flocking Control (QRIFC) [21] is proposed for collaborative UAV missions to minimize delay and energy consumption. However, cooperative UAV missions may lead to significant overhead due to the exchange of control messages among the UAVs. A solution to this challenge is to implement Distributed Constraint Optimization algorithms within multi-agent reinforcement learning frameworks [22]. These algorithms can dynamically identify the most beneficial coordination set, determining which agents to collaborate with across different scenarios. This approach reduces communication overhead, maintains overall learning performance, and ensures that system efficiency remains largely intact.

On the other hand, in MDPs, the action space is exponential in the number of UAVs, making this approach impractical in most scenarios; if there are m UAVs, each capable of taking n actions, the action space is n^m , demanding substantial computational resources and increasing the risk of sub-optimal solutions [23]. Alternative approaches to this problem have employed local optimization for the different agents, either via reward/value sharing [24] or direct policy search [25]. Reward/value sharing entails agents exchanging their learned rewards or value functions to enhance collective decision-making and coordination. Direct policy search, on the other hand, bypasses the value function approximation by directly searching for the optimal policy, allowing agents to optimize their actions based on observed outcomes individually. Additionally, value function approximation combined with message-passing schemes offers a promising solution [26]. By decomposing the overall utility function into more manageable local utility functions, these approaches facilitate coordination among agents and allow for effective routing decisions using only local information. This method ensures that the action space remains manageable by focusing solely on the UAVs pertinent to the considered local utilities, ultimately enhancing overall network performance.

The CQMR algorithm builds on the Improved Q-learning-based Multi-hop Routing (IQMR) algorithm [18]. While IQMR focuses on energy efficiency, reliable connectivity, and avoiding collisions without predefined paths, CQMR introduces the consideration of expected hops to the destination when selecting the next-hop node. Additionally, CQMR enhances performance by incorporating structured communication and coordination among UAVs.

On the other hand, the existing Q-learning-based protocols focus primarily on a single optimization criterion. In contrast, CQMR effectively balances multiple factors, which is crucial for the operational efficiency of UAV networks. Unlike traditional Q-learning protocols that rely on fixed parameters, CQMR employs adaptive adjustments to its Q-learning parameters, allowing it to respond more effectively to changing network conditions and thereby significantly enhancing overall routing performance. Most importantly, CQMR incorporates coordinated joint action selection, enabling UAVs to collaboratively evaluate and choose optimal routing actions based on the overall network environment. This collaborative assessment dramatically improves decision-making, resulting in a more resilient and efficient routing strategy in dynamic scenarios.

To the best of our knowledge, this is the first time a routing algorithm incorporates UAV coordination for data routing while simultaneously considering a multitude of network parameters for UAV network management. These advanced capabilities make CQMR an indispensable solution for ensuring robust and adaptive performance in challenging and dynamic environments. Table 1 provides a comparison of CQMR with other existing protocols.

Table 1
Comparison of Q-learning based protocols.

Protocol	Parameters									
	Coverage constraint	Collision avoidance	Energy efficiency	Packet ackn- -nowledgements	Potential network fragmentation	Expected hops	Adaptive learning rate	Adaptive discount factor	Adaptive hello interval	Coordination
QMR [6]	✗	✗	✓	✗	✗	✗	✓	✓	✓	✗
Q-FANET [9]	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
QTAR [8]	✗	✗	✓	✗	✗	✗	✓	✓	✓	✗
TARRAQ [13]	✗	✗	✓	✗	✗	✗	✓	✓	✓	✗
QGeo [10]	✗	✗	✗	✗	✗	✗	✓	✓	✓	✗
QLGR [14]	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
TONGPRS [15]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
QSRP [16]	✓	✗	✗	✓	✗	✓	✗	✗	✓	✗
QLCLRP [7]	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗
3DQ [17]	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
IQMR [18]	✓	✓	✓	✓	✓	✗	✓	✓	✓	✗
CQMR (proposed)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

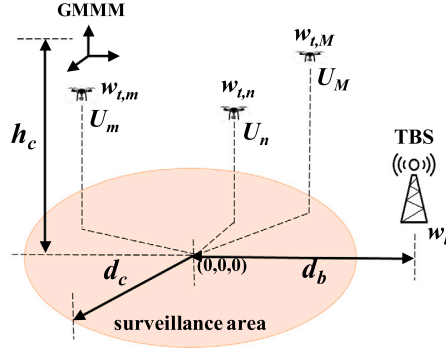


Fig. 3. Distribution of UAVs in 3D cylindrical surveillance space and TBS.

Table 2

List of parameters.

Parameter	Significance
M, U_m	Set of UAVs and their respective identifiers, where $m \in M$
d_c, h_c	Radius and height of the 3D cylindrical surveillance area.
ψ	Intensity of UAVs distributed according to a Poisson Point Process (PPP) in $\mathbb{R}^3$.
$w_{t,m}, w_b$	Locations of UAV m at time t and location of TBS
d_b	Distance between origin of 3D cylindrical surveillance space to TBS
$[v_{min}, v_{max}]$	Operating speed range of UAV
$l(r_{mn}, h_m, p)$	Path loss function based on inter-UAV distance r_{mn} , height h_m , path loss exponent p
k, K	Fading parameter and Rician Factor
SIR	Signal to Interference Ratio
g	Channel power fading gain
N_T	Neighbour Table
$\mathbb{S}(R_i, \theta)$	Sector defined by radius R_i and spanning angle θ
$\mathbf{S}, \mathbf{A}$	Set of state variables and possible actions
$\mathbf{s}, \mathbf{a}$	Specific state vector and action vector corresponding to each state/action variable in $\mathbf{S}/\mathbf{A}$ resp.
S, A	A single state variable and possible action
s, a	A specific value of a state variable S and a chosen action A .
R	Reward space
$[e, \hat{p}, \bar{p}, \bar{p}, N]$	[residual energy, packet reception status, probability of coverage, probability of collision, network outage]
$\mathbb{E}[h]$	Expected number of hops from a source UAV to the TBS
P_{L2}, P_{L3}	Packets transmitted at Layer 2 and Layer 3, resp.
A_{L2}, A_{L3}	Acknowledgements received at Layer 2 and Layer 3, resp.
θ	Minimum required SIR
r_A	Radius of the collision cross-sectional area
$\xi^{(x)}, \xi^{(y)}$	Trajectory divergence in x and y directions resp.
R_t	UAV transmission range
Q_m, Q	Utility function of UAV m and the overall system utility, resp.
f_m	Contribution of UAV m to the overall system
α, λ	Learning rates
γ	Discount factor
$e(\cdot)$	Eligibility trace
I_{xy}	Identity indicator function

3. System model

The system involves a multi-hop UAV relay network of M low-altitude UAVs and a Terrestrial Base Station (TBS) within a three-dimensional cylindrical surveillance space of radius d_c and height h_c , as seen in Fig. 3. The UAVs are Poisson Point Process (PPP) $\in \mathbb{R}^3$ distributed with intensity ψ [27]. The location of UAV m at time t is denoted as $w_{t,m}$, $0 \leq t \leq T$, while the TBS remains stationary at distance d_b from the origin of the cylindrical space at w_b . The UAVs utilize the Gauss Markov Mobility Model (GMMM) [28] for spatial positioning and operate at speeds in the range $[v_{max}, v_{min}]$.

The communication model involves a combination of distance-based path loss attenuation and small-scale fading wireless channel. Standard power-law path loss model is utilized. The path loss function is defined as $l(r_{mn}, h_m, p) = (r_{mn}^2 + h_m^2)^{-p/2}$, where r_{mn} represents the horizontal distance between transmitting UAV m and receiving UAV n , h_m is the operating height of UAV m , and p is the path loss exponent [29]. Additionally, the small-scale fading channel is modelled using Nakagami- k with a fading parameter k , computed as $k \triangleq 2(K+1)/(2K+1)$, where K represents the Rician K -factor characterizing the ratio between direct and scattered path powers. The channel power fading gain g_{mn} follows a Gamma distribution $\Gamma(k, \frac{1}{k})$ [30]. In an interference limited scenario [31], the

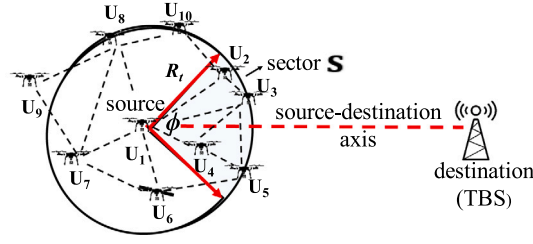


Fig. 4. Multi-UAV network with source UAV (U_1), *CoordinateNeighbourSet* (U_2, U_3, U_4, U_5), other one-hop neighbour UAVs (U_6, U_7, U_8, U_9) and non-neighbour UAV (U_{10}).

Signal to Interference Ratio (SIR) is defined as,

$$SIR_{mn} \triangleq \frac{g_{mn}l(r_{mn}, h_m)}{I} \quad (1)$$

where I is the aggregate interference power defined by $I \triangleq \sum_{z \in \mathcal{M} \setminus m} g_{zn}l(r_{zn}, h_z)$.

4. Coordinated Q-learning-based multi-hop routing Algorithm

This section presents the proposed novel Coordinated Q-learning-based Multi-hop Routing (CQMR) algorithm. It includes (i) identifying the coordination set, (ii) designing the state, action, and reward space, (iii) selecting joint action, and (iv) coordinated Q learning framework.

4.1. Identifying coordination set

The neighbour discovery module (*Algorithm 1*), constructs and updates the network topology by maintaining a *NeighbourTable* (N_T). Neighbour discovery involves exchanging *HelloMessage* over every *HelloInterval*. Each *HelloMessage* includes UAV location, residual energy, packet reception status, learning rates, discount factor, and Q-value. Upon receiving the *HelloMessage*, the UAV establishes its one-hop neighbour set, which includes nodes from which the UAV directly gets the message. A subset of one-hop UAVs, confined to a sector S , form the *CoordinateNeighbourSet* (C). Considering only the UAVs within sector S (with radius R_l and spanning an angle ϕ around the axis connecting the transmitting UAV (source node) and the destination node (TBS)) ensures targeted data transmission towards the intended destination [27]. Fig. 4 depicts the candidate neighbour, one-hop neighbour and non-neighbour UAVs. The *NeighbourTable* stores information retrieved from the *HelloMessage* sent by the *CoordinateNeighbourSet*. The *HelloInterval* dynamically adjusts according to fluctuating network conditions, ensuring timely updates while minimizing unnecessary overhead [18].

4.2. Design of state, action and reward

Each UAV operates in a dynamic environment represented by a set of state variables $\mathbf{S} = [S_1, \dots, S_n]$, and selects an action from a finite set of possible actions $\mathbf{A} = [A_1, \dots, A_g]$. A specific state s is a vector of values s_m for each state variable S_m . Similarly, a specific action a is a vector of actions a_m chosen from $\text{Domain}(A_m)$. This section details the state, action, and reward space of the proposed CQMR algorithm.

4.2.1. State

At time t , the network state is given as $s_{m,t} = [e_{m,t}, \hat{p}_{m,t}, \bar{p}_{m,t}, \bar{p}_{m,t}, N_{m,t}, \mathbb{E}_{m,t}[h]]$. The vectors $e_{m,t}, \hat{p}_{m,t}, \bar{p}_{m,t}, \bar{p}_{m,t}, N_{m,t}, \mathbb{E}_{m,t}[h]$, represent the residual energy, packet reception status, probability of coverage, probability of collision, network outage and expected hops to destination in their normalized form for the set of M UAVs ($\forall m \in M$) respectively. Normalization is achieved following the relation,

$$\hat{x} = \frac{x_{actual} - x_{minimum}}{x_{maximum} - x_{minimum}} \quad (2)$$

where x indicates the parameter to be normalized. The state space includes,

- Residual energy (e) that reflects the remaining energy levels in UAVs logged from their battery management system. Accurately tracking residual energy ensures continuous operation by preventing UAVs from prematurely stopping tasks due to power depletion.

Algorithm 1 : Neighbour Table Construction**Input:** Location of M UAVs and TBS, *HelloMessage* and sector $\mathbb{S}(R_t, \phi)$.**Output:** Updated *NeighbourTable* (N_T)**Initialization:** Set UAV identification number as $U_i = [U_1, \dots, U_M]$ **def neighbour_discovery():**# Broadcast *HelloMessage* to one-hop neighbours1: **for** for each UAV i in U_i **do**2: broadcast_hello_message(U_i)3: **end for**# Construct and update *NeighborTable* (N_T)1: **for** each received *HelloMessage* **do**2: originator_id = get_uav_id(*HelloMessage*)3: originator_location = get_uav_location(*HelloMessage*)4: **if** originator_location in $\mathbb{S}$ **then**5: $\mathbb{C}$.add(originator_id) # Add originator to Coordinate Neighbour Set6: **end if**7: **if** originator_id in N_T **then**

8: update_existing_record(originator_id)

9: **else**

10: add_new_record(originator_id)

11: **end if**12: **end for**return *CoordinateNeighbourSet* ($\mathbb{C}$)update *HelloInterval* ()

- The packet reception status $\hat{p}$ indicates whether packets have been successfully received at the TBS and thus, providing real-time insight into the current state of packet transmission. Considering a three-layer network architecture (physical layer, MAC layer, and network layer), the packet transmission success at both layers 2 and 3 is evaluated. Thus, $\hat{p}$ at time instant t is determined by the expressions involving cumulative counts of layer 2 and 3 packets transmitted (P_{m,t,L_2} and P_{m,t,L_3}) by UAV m and layer 2 and 3 acknowledgements received (A_{m,t,L_2} and A_{m,t,L_3}) by it corresponding to the packets it has transmitted up to the t th iteration of data transfer as given below, [18]

$$\hat{p}_{m,t} = \sum_{i=0}^T \frac{A_{m,t,L_2}}{P_{m,t,L_2}} + \frac{A_{m,t,L_3}}{P_{m,t,L_3}} \quad \forall m \in M \quad (3)$$

- Probability of coverage ($\hat{p}$) is the minimum SIR required for reliable data transmission along the link connecting source UAV m , and its neighbour n at time t defined as [27],

$$\hat{p}_{m,t} = \mathbb{E}[SIR_{mn}(t) \geq \theta] \quad (4)$$

where θ is the minimum SIR requirement. Estimating $\hat{p}$ allows us to ensure reliable data transmission by selecting links with adequate signal strength.

- Probability of collision $\tilde{p}$ represents the likelihood of collision at instant t between two approaching UAVs given as [27],

$$\tilde{p}_{m,t} = 1 - \exp\left(\frac{-r_{A,mn}^2}{2\tilde{\xi}_{m,t}^{(x)}\tilde{\xi}_{m,t}^{(y)}}\right) \quad (5)$$

The circular collision cross-sectional area as seen in Fig. 5 (with a radius of r_A) represents the shared space where two approaching UAVs might intersect, marking the potential collision zone. The source UAV m is assumed to lie at the centre, and its neighbour n is on the circle's circumference; hence, $r_{A,mn}$ equals the inter UAV distance r_{mn} . Therefore, in the calculation of $\tilde{p}$, r_{mn} may conveniently substitute $r_{A,mn}$. $\xi^{(x)}$ and $\xi^{(y)}$ are the divergence in the UAV trajectory from its intended path in x and y directions, respectively [27]. Understanding $\tilde{p}$ aids in UAV movement, reducing the risk of communication disruption.

- Network outage ($N_{m,t}$) indicates the number of times the UAV m has been disconnected from the network up until time t . Penalizing for N discourages actions that lead to fragmentation and encourages the learning algorithm to prioritize actions that maintain network connectivity and cohesion among UAVs.
- The expected number of hops to the destination ($\mathbb{E}_{m,t}[h]$) represents the anticipated count of intermediary nodes traversed on the path to reach the TBS. Incorporating the expected number of hops to the destination into the state space allows the routing algorithm to account for the potential path lengths and network topology variations, aiding in more informed and efficient routing decisions. $\mathbb{E}_{m,t}[h]$ is expressed as,

$$\mathbb{E}_{m,t}[h] = \sum_{h=1}^j h \mathbb{P}_{m,t}(h) \quad (6)$$

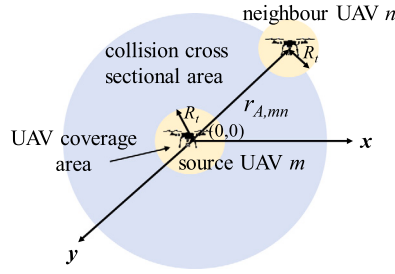


Fig. 5. Representation of the collision cross-sectional area (blue region) of the reference UAV m and its neighbour n (UAV coverage areas in peach).

where $\mathbb{P}_{m,t}(h)$ represents the probability of finding h hops between the source UAV (m) and destination (TBS) and j is the maximum hop ([Appendix](#)).

4.2.2. Action

The action of the UAV involves selecting the next hop node. Therefore, at time t , the action space is defined as $\mathbf{a}_{m,t} = [a_{m,t}^{(h)}]$ where $a_{m,t}^{(h)}$ represents the next hop UAV. The UAVs collaboratively choose the next hop $a_{m,t}^{(h)} = m_{t+1}$ to maximize the overall system utility by minimizing multi-hop transmissions and inter-UAV collisions while enhancing connectivity to ensure reliable data dissemination.

4.2.3. Reward

The reward function assigns values to actions based on the corresponding states. The reward space at time t is formulated as a joint metric of the normalized parameters in the state space as,

$$\hat{R}(s_t, a_t) = w_1 \mathbb{E}_t[h] + w_2(1 - \hat{p}_t) + w_3(1 - N_t) + w_4 \hat{p}_t + w_5 \hat{p}_t + w_6 e_t \quad (7)$$

where, $w_1 \gg w_2 \gg w_3 \gg w_4 \gg w_5 \gg w_6$, the weights are strategically assigned based on the criticality of each system parameter in optimizing data routing. The expected number of hops to the destination ($\mathbb{E}[h]$) is prioritized highest due to its substantial impact on routing efficiency, as it directly influences the minimization of multi-hop transmissions and the mitigation of packet loss risks. The inter-UAV collision probability ($\hat{p}$) is weighted next, given its importance in ensuring UAV safety and avoiding in-flight collisions. Following this, network outage (N) is considered, as maintaining network stability and avoiding disconnections are vital for sustained communication. Packet reception status ($\hat{p}$) is subsequently considered to monitor the integrity of ongoing data transfers. However, N is prioritized over $\hat{p}$ due to its significant role in preserving overall network connectivity. The probability of coverage ($\hat{p}$) is then factored in to ensure the reliability of the communication link. Lastly, residual energy (e) is considered. This is because, without a stable communication link, data transmission may fail, regardless of the remaining energy reserves. Consequently, $\hat{p}$ is emphasized over e . The joint reward function is defined as,

$$R(s_t, a_t) = \begin{cases} R_{max}, & \text{next hop is TBS} \\ R_{min}, & \text{next hop is local minima} \\ \hat{R}, & \text{otherwise} \end{cases} \quad (8)$$

According to (8), the transmitting UAV receives the highest reward when the next hop node is the TBS. Conversely, suppose the routing algorithm is trapped in a sub-optimal solution called a local minimum the transmitting UAV receives the minimum reward. Otherwise, the reward follows the criteria outlined in Eq. (7).

4.3. Joint action selection

This section presents a model for joint action selection among multiple UAVs to maximize the sum of their local utilities. The block diagram in [Fig. 6](#) illustrates the mechanism of coordinated Q-learning-based joint action selection.

For UAVs operating within the state space $\mathbf{S} = [S_1, \dots, S_n]$ and selecting actions from a finite set $\mathbf{A} = [A_1, \dots, A_g]$, the overall utility (Q) depends on all state variables $\mathbf{S}$ and the collective actions taken by all UAVs $\mathbf{A}$. Therefore, determining the action that maximizes Q is challenging, as it requires knowledge of the joint actions of all UAVs. A pragmatic approach to this issue involves approximating Q as a sum of individual local utilities Q_m , expressed as $Q = \sum_m Q_m$ [32]. Here, Q_m is a function of the state variables relevant to UAV m and the actions it takes. Importantly, Q_m has a local structure, meaning each UAV's utility depends only on a specific subset of state variables and actions, rather than on the entire state and action space of all UAVs. By exploiting this local structure using a coordination graph, one can determine the optimal joint action with minimal communication between UAVs, limited to their respective *CoordinationNeighbourSet*. This approach simplifies the joint action selection problem by focusing on local interactions.

The set of state and action variables that affect Q_m is defined as $Scope[Q_m] \subset S \subset A$. The value $Q_m(s, a)$ is determined when the variables within $Scope[Q_m]$ are assigned specific values from s, a . UAV m contributes to a part of Q_m , which depends on the state

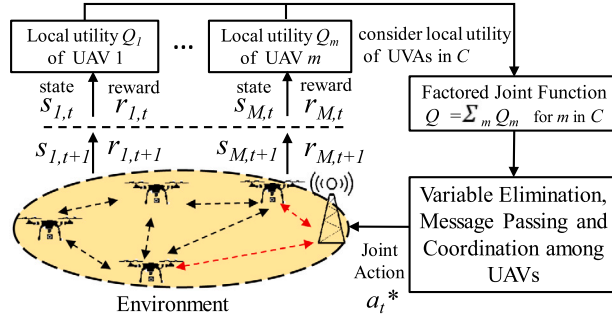


Fig. 6. Illustration of Multi-UAV joint action selection mechanism using coordinated Q-learning.

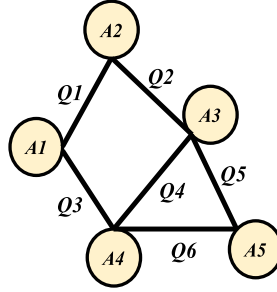


Fig. 7. Coordination graph representing a source UAV and three *CoordinateNeighbourSet* UAVs.

$s = [s_1, \dots, s_m]$, assuming that all relevant state variables are fully observable. Each Q_m is solely determined by an action A_m taken by UAV m . However, because Q_m also relies on the actions of other UAVs, coordination among the UAVs is essential to maximize the overall Q-value. To represent this requirement, a coordination graph is used where, each node represents a UAV, and an edge exists between two UAVs if they need to coordinate their actions to directly optimize a specific Q_m .

Consider the network topology depicted in Fig. 4, where U_1 serves as the source UAV, while U_2, U_3, U_4 , and U_5 form part of its *CoordinateNeighbourSet*. U_1 collaborates with its *CoordinateNeighbourSet* to determine the subsequent hop for forwarding data towards the TBS. Fig. 7 illustrates the coordination graph for the network depicted in Fig. 4. In the coordination graph, A_m represents the action that UAV m (denoted as U_m) can take, and Q_m represents its local utility. In this context, $Q_m(a_m, a_n)$ signifies the utility of UAV m when taking an action a_m given the action a_n of another UAV n , contributing to the overall utility Q .

The overall utility of the considered network is expressed as a linear combination of local utilities as,

$$Q = \max_{a_1, a_2, a_3, a_4, a_5} Q_1(a_1, a_2) + Q_2(a_2, a_3) + Q_3(a_1, a_4) + Q_4(a_3, a_4) + Q_5(a_3, a_5) + Q_6(a_4, a_5) \quad (9)$$

The coordination graph is solved using the variable elimination method, which involves maximizing one variable at a time instead of maximizing over the sum of local utilities. During this process, the maximized variable at each step is eliminated, as its influence has already been considered in optimization and does not contribute additional value.

Beginning the optimization with U_5 , which requires knowledge of the values of A_3 and A_4 . Thus (9) transform to (10) as,

$$Q = \max_{a_1, a_2, a_3, a_4} Q_1(a_1, a_2) + Q_2(a_2, a_3) + Q_3(a_1, a_4) + Q_4(a_3, a_4) + \max_{a_5} [Q_5(a_3, a_5) + Q_6(a_4, a_5)] \quad (10)$$

A conditional strategy to calculate the value f_5 , which represents the contribution U_5 makes to the system based on different action choices of UAVs U_3 and U_4 is determined. Specifically, $f_5(A_3, A_4)$ at a_3, a_4 is given by $\max_{a_5} [Q_5(a_3, a_5) + Q_6(a_4, a_5)]$, reflecting the combined effect of A_3 and A_4 on the system while assuming that U_5 optimizes its decision in response to the actions of U_3 and U_4 .

Thus, Eq. (10) reduces to computing,

$$Q = \max_{a_1, a_2, a_3, a_4} Q_1(a_1, a_2) + Q_2(a_2, a_3) + Q_3(a_1, a_4) + Q_4(a_3, a_4) + f_5(a_3, a_4) \quad (11)$$

now there is one fewer UAV. Next U_4 makes its decision:

$$\max_{a_1, a_2, a_3} Q_1(a_1, a_2) + Q_2(a_2, a_3) + f_4(a_1, a_3) \quad (12)$$

where, $f_4(a_1, a_3) = \max_{a_4} [Q_3(a_1, a_4) + Q_4(a_3, a_4) + f_5(a_3, a_4)]$.

Subsequently U_3 makes its decision:

$$\max_{a_1, a_2} Q_1(a_1, a_2) + f_3(a_1, a_2) \quad (13)$$

where, $f_3(a_1, a_2) = \max_{a_3} [Q_2(a_2, a_3) + f_4(a_1, a_3)]$.

Following that U_2 makes its decision:

$$f_2(a_1) = \max_{a_2} Q_1(a_1, a_2) + f_3(a_1, a_2) \quad (14)$$

U_1 then chooses an action a_1 that maximizes $f_1 = \max_{a_1} f_2(a_1)$.

Further, the maximizing set of actions by performing the entire sequence in reverse is obtained. Initially, the optimal choice for f_1 selects an action a_1^* for U_1 . In response to this commitment, U_2 selects action a_2^* that maximizes $f_2(a_1^*)$. As a result, U_3 , then U_4 , and subsequently U_5 are compelled to choose their actions accordingly. In this scenario, not every UAV is required to coordinate directly with all other UAVs, thus simplifying the coordination process.

Algorithm 2 determines the best set of actions for a group of UAVs based on their dependencies, represented as a coordination graph. The algorithm maintains a set F of functions initialized with $(Q_1, \dots, Q_g)$. The process begins by iteratively eliminating UAVs, selecting an uneliminated UAV U_i , and combining relevant utility functions that include U_i in their scope. A new function f is created by maximizing over the actions of U_i and summing the utility contributions, then added back to the set of functions F . This continues until all UAVs are eliminated. In the second phase, the algorithm recovers the maximizing set of actions by starting with the last function and iteratively determining the best action choices for each UAV, updating the utility functions accordingly to reflect these choices. The result is the optimal joint action set for all UAVs, maximizing the overall utility.

Note that the joint action taken by UAVs U_1, U_2, U_3, U_4 , and U_5 , with U_1 as the source node, remains consistent with their actions under different source node scenarios. This coherence is maintained through structured optimization processes and conditional decision-making strategies, aligning actions with specific contexts and avoiding contradictions in network coordination. Furthermore, the coordination graph representation elucidates the direct coordination required for optimizing specific utilities, ensuring that joint actions are in line with the overarching goal of maximizing total utility while sustaining consistency in UAV interactions.

Algorithm 2 : Joint Action Selection

Input: Set of functions $(Q_1, \dots, Q_g)$ and coordination graph representing dependencies between UAVs

Output: Joint Action Selection

Initialization: Initialize set F of functions to $(Q_1, \dots, Q_g)$

def joint_action_selection():

Eliminate UAV U_i

```

1: while there are uneliminated_UAV  $U_i$  do
2:   select uneliminated_UAV  $U_i$ 
3:    $\forall$  functions  $f_1, \dots, f_L \in F$  whose scope contains  $U_i$ 
4:   define new_function  $f = \max_{a_i} \sum_k f_k$  and add it to  $F$ 
5:   mark  $U_i$  as eliminated
6: end while

```

Recover maximizing set of actions

```

1: find  $f_1 = \max_{a_i} f_i$ .
2: for  $f_k$  from 1,  $\dots$ ,  $g$  do
3:   find maximizing_action_choice  $a_k^*$  given the previous function  $f_{k-1}$ 
4:   update  $f_k = \max_{a_i} f_k(a_k^*)$ 
5: end for

```

4.4. Coordinated $Q(\lambda)$ -learning framework

In the coordinated $Q(\lambda)$ -learning framework, UAVs operate under the principles of Q-learning, where they observe the environment state s_t at time t , take action a_t receive reward r_t , and transition to the next state s_{t+1} at time $t+1$, thereby learning the state action pair from observation of the quadruples (s_t, a_t, r_t, s_{t+1}) . The objective is to determine an optimal policy π , which maps $\pi : s \rightarrow a$ with $\pi(s)$ being the optimal policy taken in state $s \in \mathcal{S}$. An optimal policy π^* maximizes the expected discounted reward as,

$$\pi^* = \arg \max_{\pi} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(s_t)) \right] \quad (15)$$

where, γ , is the discount factor.

The policy π guides the agent's decision-making process by determining the actions to take in different states, while the Bellman equation is utilized to calculate the expected discounted cumulative rewards associated with these actions, thereby updating the agent's Q-values iteratively as given in (16).

$$Q(s_t, a_t) = \mathbb{E}_\pi \left[\sum_{i=0}^N \gamma^i R(t+i+1) \middle| s_t, a_t \right] e_t(s, a) \quad (16)$$

where, $e(s, a)$ [33] is the eligibility trace. The eligibility trace reinforces the recently revisited state-action pairs (s, a) , emphasizing their importance. $e(s, a)$ is defined as,

$$e_t(s, a) = I_{ss_t} I_{aa_t} + \begin{cases} \gamma \lambda e_{t-1}(s, a), & a_t = \pi_{t-1}^*(s_t) \\ 0, & \text{otherwise} \end{cases} \quad (17)$$

where I_{xy} is an identity indicator function which equals to 1 if $x = y$ and 0, otherwise. λ is a learning parameter. Eq. (16) is expressed as

$$\hat{Q}(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [R_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)] e_t(s, a) \quad (18)$$

where α and γ are the learning rate and discount factor, both varying within the range of (0,1). α governs the rate at which Q-values are updated, while γ weighs the significance of immediate versus future rewards. $Q(s_t, a_t)$ and $\hat{Q}(s_t, a_t)$ denotes the previous and updated Q-value respectively. The term $\max_a Q(s_{t+1}, a)$ estimates the optimal future value, while R_{t+1} signifies the reward obtained when transitioning from state s_t to state s_{t+1} .

The Q-learning update procedure requires only a differentiable approximation framework that is compatible with the action selection mechanism. Leveraging the advantage of flexibility in the Q-learning update technique, it is assumed that each UAV maintains a local Q_m -function defined over a subset s_m of the state variables, which can comprise the entire state s , and a subset a_m of the action variables a , including its actions and potentially the actions of other agents. Thus total utility Q is a function of global state s and the joint action vector a^* as,

$$Q(s, a^*) = \sum_{m=1}^g Q_m(s_m, a_m) \quad (19)$$

Utilizing the message passing algorithm on the coordination graph (Section 4.C) allows us to determine the maximum value $U(s_{t+1})$ for any given state s_{t+1} without requiring maximization over a large action space, owing to the Q-function's factorization as a linear combination of local Q_m functions, each dependent solely on a subset a_m of a .

R and $Q(s, a)$ are the remaining variables in the update equation that need to be determined. The reward is observed directly, while the Q-value is calculated through message passing akin to the one in the coordination graph, with the action of each UAV remaining fixed according to the assignment in a .

The parameters α and γ are dynamic, adjusting in response to changing network conditions to ensure adaptability and optimal performance [18]. Unlike α , λ remains constant to preserve consistency in memory trace, ensuring the UAV relies on a dependable record of past experiences.

4.5. Working of CQMR

The CQMR algorithm functions in two phases: exploration and exploitation. During exploration, it adopts an episodic structure with a set episode length T . Each episode initiates with UAVs distributed in a specific area according to $PPP \in \mathbb{R}^3$, while the TBS remains static. UAVs actively transmit data packets towards the TBS, employing an ϵ -greedy policy, thus preventing the algorithm from local optima. In the exploitation phase, the algorithm operates until convergence, which is defined as the next hop node being, the TBS. UAVs take joint action based on learned policies, leading to state transitions and reward outcomes and aiming to maximize the global utility of the system using the functions of joint action selection, convergence checks, and Q-value update.

Algorithm 3, the Coordinated Q-learning-based Multi-Hop Routing Algorithm, begins by initializing parameters such as the Q-matrix, exploration rate, learning rate, and episode length, among others. It then formulates the neighbour set for each UAV and proceeds to the exploration phase. During exploration, each UAV performs actions based on an ϵ -greedy policy, observes state variables and rewards, and updates the Q-matrix accordingly. This phase continues until the episode length is reached. In the exploitation phase, the algorithm iteratively refines its routing decisions, balancing exploration and exploitation through joint action selection and convergence checks. The algorithm aims to find optimal routing policies for UAVs, promoting efficient communication in multi-hop networks while adapting to changing network conditions

4.6. Time complexity analysis

4.6.1. The time complexity analysis of Algorithm 1

Initially, each UAV broadcasts a HelloMessage. Following this, the operations of fetching the positions of the originator UAV and checking if each of them is in the neighbour table involve iterating through the messages received. In the worst-case scenario, when

Algorithm 3 : Coordinated Q-learning-based Multi-Hop Routing Algorithm**Input:** Location of M UAVs and TBS,**Output:** Routing policy (π^*) for each UAV-TBS pair.**Initialization:** Set Q-matrix with random values, exploration rate (ϵ) learning rate (α), discount factor (γ), episode length (T), state_variables() and reward.# Formulation of *CoordinateNeighbourSet()*

```

1: for each UAV  $m$  do
2:   call Algorithm 1
3: end for
# Exploitation Phase
1: for each time step  $t$  from 1 to episode_length  $T$  do
2:   for each UAV  $m \in M$  do
3:     perform  $\epsilon$ -greedy_action
4:     observe next_state_variables(), reward
5:     update Q-matrix()
6:     update state_variables()
7:     if episode ends then
8:       break
9:     end if
10:   end for
11: end for

```

Exploration Phase

```

1: while not converged do
2:   for each UAV  $m$  do
3:     if random_value <  $\epsilon$  then
4:       action = random_action()
5:     else
6:       action = joint_action_selection()
7:     end if
8:     observe next_state_variables(), reward
9:     update Q-matrix()
10:    update state_variables()
11:   end for
12:   converged = check_convergence()
13: end while

```

Functions

function joint_action_selection()

```

1: call Algorithm 2
2: return maximizing_action_choice

```

function check_convergence()

```

1: check if next_hop = TBS
2: return convergence_status

```

each UAV receives messages from all others, the size of the neighbour table equals M , resulting in a time complexity of $O(M^2)$.

4.6.2. The time complexity analysis of Algorithm 2

The joint action selection algorithm determines the best-coordinated actions for all UAVs. In the worst-case scenario, when each UAV may need to interact with every other UAV to evaluate and combine functions, the time complexity is $O(M^2)$.

4.6.3. The time complexity analysis of Algorithm 3

The exploration phase runs for a specified number of time steps T . During each time step, the algorithm processes all M UAVs. This processing involves selecting actions, observing states and rewards, updating the Q-matrix, and updating state variables. Consequently, the exploration phase has a time complexity of $O(MT)$. On the other hand, the exploitation phase iterates until convergence, denoted by I . In each iteration, the algorithm processes all M UAVs similar to the exploration phase. Thus, the time complexity of the exploitation phase is $O(IM)$.

Therefore the overall time complexity of the CQMR algorithm, given $M \ll T$, is $O((T + I)M)$.

5. Results and discussion

This section presents the performance analysis of the proposed CQMR algorithm through extensive simulations. First, the expected number of hops required to reach the destination (TBS), considering the impact of varying system parameters such as the distance to the destination and the number of UAVs is investigated. Second, the system's robustness to sudden disruptions, including (i) signal strength degradation, (ii) UAV collisions, and (iii) UAVs temporarily leaving the network for recharging and then successfully reintegrating into the network is evaluated. Finally, the CQMR algorithm is compared with the existing IQMR algorithm [18], underscoring the improvements in expected hops to destination, energy, and successful data transmission efficiency. CQMR builds upon and enhances IQMR. While both algorithms share similar state, action, and reward formulations, their learning approaches differ substantially: the CQMR algorithm leverages coordinated $Q(\lambda)$ -learning, whereas the IQMR algorithm employs independent $Q(\lambda)$ -learning. The comparison is intentionally focused on IQMR because it serves as a strong and relevant baseline—being our previously published algorithm that already outperforms several state-of-the-art routing techniques in terms of energy efficiency and reliability. Comparing CQMR against IQMR, isolates and quantifies the specific performance gains introduced by coordination among agents. This focused comparison helps in directly evaluating the added value of coordination in multi-agent reinforcement learning settings without confounding effects from other algorithmic differences that would arise if unrelated methods were used as baselines. Broader bench-marking against other techniques was already conducted and reported in the prior work on IQMR.

The simulation environment includes UAVs distributed in a 3D space with a radius of 1000 m and a maximum altitude of 300 m. The UAVs operate at altitudes between 100 and 300 m, with speeds ranging from 10 to 30 m/s. Each UAV has a uniform transmission power of 1 W and a radio range of 250 m. In each iteration, UAVs can transmit data directly to the terrestrial base station (TBS) or nearby UAVs within their coverage range. Additionally, each UAV is equipped with an energy capacity of 207,792 J (equivalent to 11.1 V and 5200 mAh). Table 2 provides a detailed summary of the parameters defining the operational boundaries and capabilities of the UAVs in the simulation scenario.

5.1. Analysing expected number of hops to the destination

This section examines how system parameters, such as the distance to the destination and the number of UAVs, influence the expected number of hops to reach the destination.

Fig. 8 displays the mean hop counts for different distance intervals from a source node to the destination node, with the x -axis representing the distance to the destination (in metres) and the y -axis indicating the mean hop count. The UAV count considered in this context is 15. When analysing the mean hop counts, CQMR shows mean hop counts of 2, 5, 6, and 8 for distances of 400, 800, 1200, and 1600 m to the destination, respectively. In comparison, IQMR exhibits mean hop counts of 3, 7, 9, and 11 for the same distance ranges. The percentage difference in the mean hop count between IQMR and CQMR is also represented in the graph. The figure indicates that hop counts increase with greater distances, reflecting the need for more intermediate nodes to relay data over longer distances. The figure also highlights CQMR's superior efficiency. On average, IQMR requires approximately 44.37% more hops than CQMR for the same distance interval.

Fig. 9 illustrates the impact of the number of UAV nodes in the network on the expected number of hops to the destination. The x -axis represents the number of UAVs, while the y -axis indicates the mean hop count. The source UAV is assumed to be located 800 m away from the TBS. Examining the mean hop counts, CQMR exhibits mean hop counts of 7, 6, 5, 5, and 5 for the number of UAVs varying as 5, 10, 15, 20, 25, and 30, respectively. In contrast, IQMR shows mean hop counts of 9, 8, 7, 7, 7, and 8 for the same node ranges. The percentage difference in the mean hop count between IQMR and CQMR is also represented in the graph. The Fig. 8 indicates that the mean hop counts initially decrease and then plateau as the network becomes sufficiently dense, with the existing UAVs providing adequate coverage. It is observed that the mean hops for IQMR slightly increase when $M = 30$. As the number of UAVs continues to rise beyond a certain point, the network may experience minor inefficiencies due to potential congestion, which leads to a slight increase in the expected hop count. The figure also shows IQMR requires approximately 40.32% more hops for the same number of UAV nodes.

5.2. System dynamics for changing system parameters

This section explores the system's response to abrupt changes in parameters and its ability to stabilize, illustrating CQMR's resilience and adaptability in dynamic environments. The sudden changes considered include (i) signal drops, (ii) UAV collisions, and (iii) UAVs temporarily exiting the network for recharging and subsequently reintegrating.

The cumulative reward graphs in Figs. 10, 11, and 12 are analysed to understand how the system behaves under abrupt changes in operating conditions and how it stabilizes. The x -axis represents the number of iterations of data transfer, while the y -axis represents the cumulative reward of the system. Fig. 10 demonstrates how the system reacts to an abrupt 5 dB decrease in signal strength. Fig. 11 depicts the system's behaviour when 10% of the operating UAVs experience physical collisions and are permanently removed from the network from the time of the collision. Fig. 12 illustrates the scenario where 15% of the operating UAVs undergo power depletion and temporarily leave the network for 400 ms to recharge. These UAVs then reintegrate into the network one after the other over a 40 ms period to prevent potential network congestion and message overhead.

From Figs. 10, 11, and 12, it is evident that CQMR achieves fewer disruptions and stabilizes more rapidly compared to IQMR. However, each of these sudden scenarios distinctly affects the overall system performance. Fig. 11 illustrates that UAV collisions have a significant impact compared to signal attenuation (Fig. 10) and UAVs temporarily departing the network (Fig. 12). Collisions

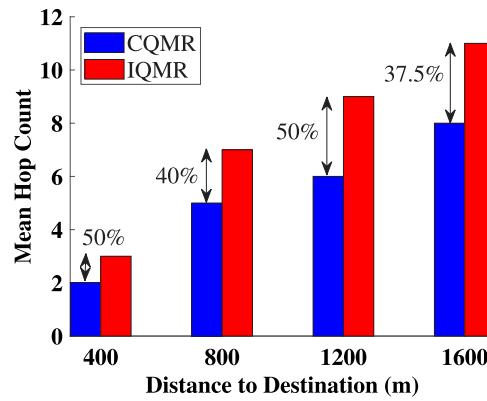


Fig. 8. Mean hop count for different distance intervals to the destination node.

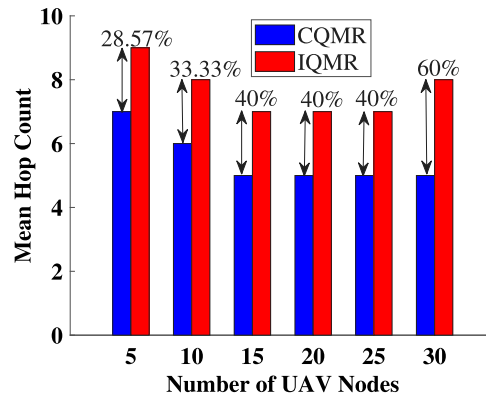


Fig. 9. Mean hop count for varying numbers of UAV nodes in the network.

Table 3

List of simulation parameters [4,8,18,27].

Application setting		Network setting	
Number of UAVs:	50	Transmission power:	1 W
Number of TBS:	1	Residual energy:	207 792 J
Surveillance area:	Cylindrical (R = 1000 m, H = 300 m)	SIR threshold:	0 dB
Transmission range:	250 m	Trajectory divergence:	3 m
UAV operation height:	100–300 m	Traffic type:	2 Mbps CBR
UAV speed:	10–30 m/s	MAC Protocol:	802.11n
Prop. model:	Nakagami small scale fading	Bandwidth	20 MHz
		Exploration time:	1000 s
		Network simulator:	MATLAB

cause permanent physical damage to UAVs, leading to costly equipment replacement, data loss, and immediate network disruptions that cause widespread communication breakdowns. In contrast, UAVs' transient exits from the network affect connectivity and route stability, disrupting communication paths and potentially causing delays as the network adjusts. Despite this, temporary exits are generally less damaging than collisions because they do not result in physical damage to the UAVs themselves or require extensive repair or replacement of equipment. On the other hand, signal loss leads to increased packet loss and re-transmissions, which can be mitigated through strategies like signal boosting, error correction, or re-routing. Thus, signal drop has a lesser impact than UAV collisions and transient exits from the network (see Table 3).

5.3. Performance comparison: CQMR with IQMR

This section compares and analyses the performance of the proposed CQMR algorithm with IQMR.

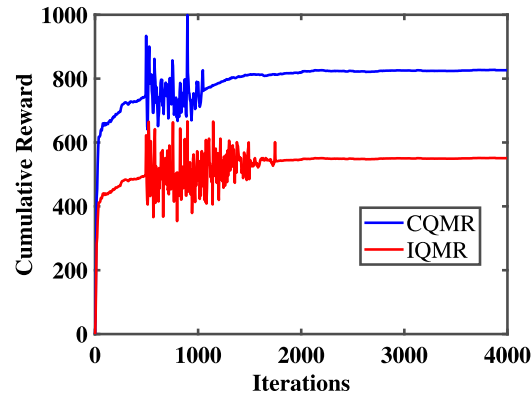


Fig. 10. The effect of sudden changes in system parameters on cumulative reward for 5 dB decrease in signal strength.

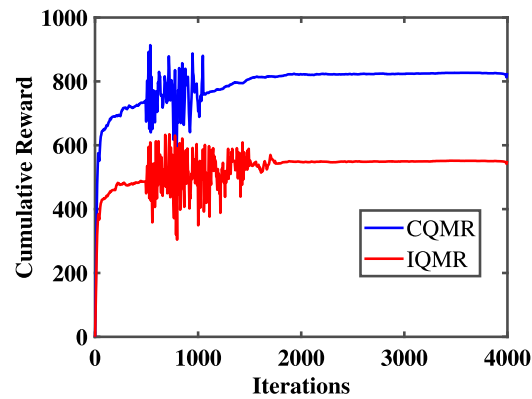


Fig. 11. The effect of sudden changes in system parameters on cumulative reward for 10% of operating UAVs experiencing collisions.

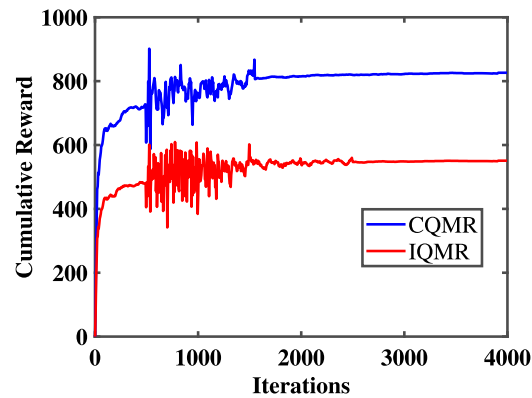


Fig. 12. The effect of sudden changes in system parameters on cumulative reward for 15% of UAVs experiencing power depletion and temporarily leaving the network for 400 ms to recharge, followed by reintegration in batches over 40 ms period.

1. *CQMR Energy Efficiency*: The residual energy of the network is the sum of the remaining energies of all the UAVs after each iteration of data transmission. Initially, the UAVs are powered with 11.1 V and 5200 mAh battery. The energy expended for data dissemination follows the distance-dependent energy consumption model described in [18]. In Fig. 13, the left y-axis shows the decline in residual energy, while the right y-axis plots the fraction of UAVs maintaining connectivity. As depicted in Fig. 13, the CQMR algorithm demonstrates superior energy consumption compared to IQMR. At the end of 8000 iterations of data transfer, CQMR retains 49.08% of the energy compared to IQMR's 36.34%, resulting in a 12.74% increase in energy efficiency for CQMR over IQMR.

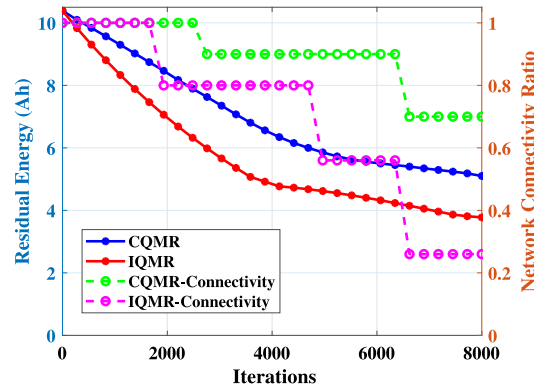


Fig. 13. Comparison of remaining energy levels and connectivity of CQMR and IQMR.

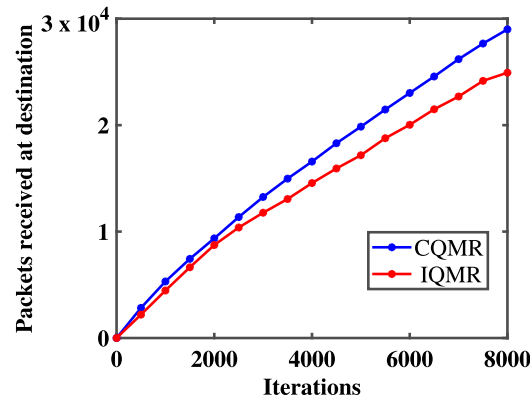


Fig. 14. Comparison of successful data packet deliveries to the destination across multiple iterations of data transmission for CQMR and IQMR.

To further analyse the impact of residual energy on network performance, it is compared against the network connectivity ratio over time. For CQMR, the network maintains full connectivity until residual energy drops to approximately 73%. Connectivity then reduces to 90% and subsequently to 70% as the energy declines further by 21%. In contrast, IQMR experiences a steeper decline: connectivity drops to 80% when residual energy falls to 68%, then to 56% after a further 23% energy drop, and finally plummets to 26% with an additional 6% reduction. These observations highlight that CQMR not only conserves energy more effectively but also sustains network connectivity longer, underscoring its advantage in maintaining reliable communication under energy-constrained conditions.

1. **CQMR Successful Data Transmission Efficiency:** Fig. 14 illustrates the data throughput, reflecting the successful transmission of packets to the TBS. Out of the total 3×10^4 transmitted packets in the network, CQMR achieves around 2.9×10^4 successful transfers to the TBS by the 8000th iteration, while IQMR achieves approximately 2.4×10^4 successful data transfers to the TBS. CQMR demonstrates a packet loss rate of approximately 3.33%, contrasting with IQMR's 16.67%, resulting in CQMR achieving a 13.34% higher success rate in data transmission.

5.4. Overhead analysis due to control message exchange

In CQMR, control-plane overhead arises from two main sources: periodic *HelloMessages* for neighbour discovery and coordination messages used during joint action selection. Below quantifying each component and then assessing their combined impact relative to data traffic and energy use.

(1) Hello Message Overhead

Every UAV broadcasts a *HelloMessages* of size 100 bytes once every second. With 50 UAVs in the network, this yields an aggregate overhead of $50 \text{ UAVs} \times 100 \text{ bytes/s} = 5000 \text{ bytes/s}$.

(2) Coordination Message Overhead

During each routing decision interval (every 0.5 s—chosen to match typical UAV movement of 10–30 m/s, which causes 5–15 m of travel per interval, and to react promptly to link changes without excessive messaging), each UAV exchanges a 100-byte coordination packet. For 50 UAVs, this yields $50 \text{ UAVs} \times 100 \text{ bytes/0.5 s} = 10000 \text{ bytes/s}$.

(3) Control-to-Data Ratio

If each UAV generates application data at rate 2 Mbps, the combined control overhead (*HelloMessage* + coordination) of 15 000 bytes/s gives a control-to-data ratio of $(5000 + 10000)/(250000) = 0.06$.

(4) Energy Overhead

Assuming a 1 W transmit power, an extra 15 000 bytes/s of control traffic consumes roughly $1 \text{ W} \times (15000)/(15000 + 250000) = 0.057$

Therefore from the above mathematical analysis it is clear that, even with 50 UAVs sending 100 bytes of *HelloMessage* every second and 100 bytes coordination packets every 0.5 s, the total control overhead is only 15 kB/s. Compared to a 2 Mbps (250 kilo bytes per second) data stream, control signalling is just 6% of the throughput. In energy terms, it consumes under 6% of a 1 W transmit budget. This confirms that CQMR's control-plane messaging is modest and easily manageable, even in large, high-rate UAV deployments.

6. Conclusion

This study presents a novel coordinated Q-learning-based CQMR algorithm for multi-hop data routing in dynamic UAV networks serving disaster management operations. To the best of our knowledge, this is the first instance of a routing algorithm that introduces a joint action selection mechanism by coordinating UAVs through value function approximation and a message-passing scheme to exchange reinforcement signals, utility functions, and conditional policies implemented as a negotiation scheme over the coordination graph for joint action selection using Q(λ) framework. Further, the proposed novel approach also offers a pragmatic solution that ensures minimal hops to the destination, enhances energy efficiency, maintains connectivity, avoids UAV-to-UAV collisions, and supports adaptive network restoration through dynamic UAV path planning.

The study investigates how changes in system parameters, such as distance to the destination, number of UAVs, SIR, and inter-UAV distance, affect the expected hops to reach the destination. It further evaluates the system's response to sudden disruptions like signal strength drops, UAV collisions, and UAVs exiting the network to recharge, followed by reintegration. In addition, CQMR is evaluated against IQMR, demonstrating CQMR's superior performance with a 12.47% improvement in energy consumption efficiency and a 13.34% increase in data transmission success rate. While on average, IQMR requires approximately 44.37% more hops than CQMR for the same distance interval and 40.32% more hops for the same number of UAV nodes in the network.

Further extensions to the CQMR algorithm include techniques for predictive analytics and extending its application to large-scale UAV networks with heterogeneous capabilities and diverse mission requirements.

CRedit authorship contribution statement

N.P. Sharvari: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dibakar Das:** Writing – review & editing, Methodology. **Jyotsna Bapat:** Writing – review & editing, Supervision, Funding acquisition. **Debabrata Das:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Expected number of hops between a source and destination node

This appendix computes the expected number of hops ($\mathbb{E}[h]$) between a source node (originator UAV) and destination node (TBS) pair.

$$\mathbb{E}[h] = \sum_{h=1}^j h \cdot \mathbb{P}(h) \quad (\text{A.1})$$

where j is the maximum hop in network, such that, $1 \leq j \leq M$.

In a $\text{PPP} \in \mathbb{R}^3$ of intensity ψ , the probability of finding k nodes in sector $\mathbb{S}$ of area A is given by,

$$\mathbb{P}[k \text{ nodes in } A] = \frac{(\psi A)^k e^{-\psi A}}{k!} \quad (\text{A.2})$$

Then the probability of not having any node in A is,

$$\mathbb{P}[k = 0 \text{ in } A] = e^{-\psi A} \quad (\text{A.3})$$

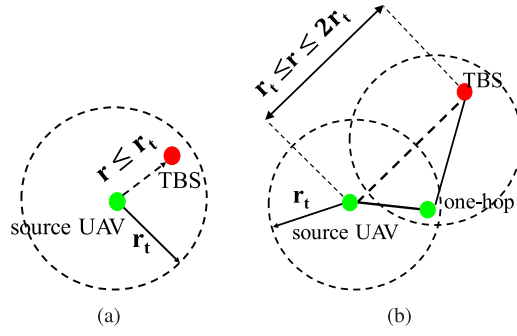


Fig. 15. Representing (a) one-hop and (b) two-hop connectivity.

Further in $\text{PPP} \in \mathbb{R}^3$ of intensity ψ , the inter-UAV distance r is a random variable in R_n . The distance R_n between a originator UAV and its n th neighbour is distributed according to the generalized Gamma distribution [34]. Thus, the Probability Density Function (PDF) of R_n is given as,

$$f_{R_n}(r) = \frac{(\psi r)^n}{r \Gamma(n)} e^{-\psi r} \quad (\text{A.4})$$

To compute $\mathbb{E}[h]$, the probability of occurrence for one-hop and two-hop connection cases are first defined. The analysis is extended to determine the likelihood for the j -hop scenario, providing a generalized expression [35].

Case 1: One hop - The probability of a one-hop connection, considering TBS to be within the transmission range R_t of the originating UAV, implying that the inter-UAV distance r satisfies $1 \leq r \leq R_t$ is derived. The same is illustrated in Fig. 14.a. In this scenario, a single data transmission hop is sufficient to reach the destination, thereby constituting a one-hop connection (see Fig. 15).

Hence, the probability of a one-hop connection denoted as $\mathbb{P}(0 < r < R_t)$, is expressed as,

$$\mathbb{P}(\text{one} - \text{hop}) = \int_0^{R_t} f_{R_n}(r) dr \quad (\text{A.5})$$

$$\mathbb{P}(\text{one} - \text{hop}) = \int_0^{R_t} \frac{(\psi r)^n}{r \Gamma(n)} e^{-\psi r} dr \quad (\text{A.6})$$

To simplify the integral in (A.6) change of variable as $t = \psi r$ is used, resulting in Eq. (A.7).

$$\mathbb{P}(\text{one} - \text{hop}) = \frac{\psi}{\Gamma(n)} \int_0^{\psi R_t} t^{n-1} e^{-t} \frac{dr}{\psi} \quad (\text{A.7})$$

(A.7) is solved using the definition of lower incomplete gamma function $\int_0^x e^{-t} t^{s-1} dt = \gamma(s, x)$ [36].

$$\mathbb{P}(\text{one} - \text{hop}) = \frac{\gamma(n, \psi R_t)}{\Gamma(n)} \quad (\text{A.8})$$

Case 2: Two hop - To determine the probability of a two-hop connection, the case where the originating UAV and the TBS are separated by a distance r such that $R_t < r < 2R_t$ is considered as seen in Fig. 14.b.

$$\mathbb{P}(\text{two} - \text{hop}) = \int_{R_t}^{2R_t} f_{R_n}(r) dr [1 - e^{-\psi A}] \quad (\text{A.9})$$

Similar to obtaining (A.8), (A.9) is solved resulting in (A.10).

$$\mathbb{P}(\text{two} - \text{hop}) = \frac{\gamma(n, \psi R_t) - \gamma(n, \psi(2R_t))}{\Gamma(n)} [1 - e^{-\psi A}] \quad (\text{A.10})$$

Using the property of generalized incomplete gamma function $\gamma(s, x) - \gamma(s, y) = \gamma(s, x, y)$ [36], (A.10) is further simplified to,

$$\mathbb{P}(\text{two} - \text{hop}) = \frac{\gamma(n, \psi R_t, \psi(2R_t))}{\Gamma(n)} [1 - e^{-\psi A}] \quad (\text{A.11})$$

By inference from the equations in (A.8) and (A.11), the generalized expression for the probability of an j -hop connection is obtained as,

$$\mathbb{P}(j - \text{hop}) = \frac{\gamma(n, \psi(j-1)R_t, \psi j R_t)}{\Gamma(n)} [1 - e^{-\psi A}]^{j-1} \quad (\text{A.12})$$

Based on the derived probabilities, $\mathbb{E}(h)$ is determined as,

$$\begin{aligned}\mathbb{E}[h] &= \sum_{h=1}^j \mathbb{P}(h) \\ &= \sum_{h=1}^j h \left[\frac{\gamma(n, \psi(h-1)R_t, \psi h R_t)}{\Gamma(n)} \right] [1 - e^{-\psi A}]^{h-1}\end{aligned}\quad (\text{A.13})$$

Data availability

Data will be made available on request.

References

- [1] G. Geraci, A. Garcia-Rodriguez, M.M. Azari, A. Lozano, M. Mezzavilla, S. Chatzinotas, Y. Chen, S. Rangan, M.D. Renzo, What will the future of UAV cellular communications be? A flight from 5G to 6G, *IEEE Commun. Surv. Tutor.* 24 (3) (2022) 1304–1335.
- [2] J. Kedys, I. Tchappi, A. Najjar, UAVs for disaster management - an exploratory review, *Procedia Comput. Sci.* 231 (2024) 129–136, 14th International Conference on Emerging Ubiquitous Systems and Pervasive Networks / 13th International Conference on Current and Future Trends of Information and Communication Technologies in Healthcare (EUSPN/ICTH 2023).
- [3] M. Mozaffari, W. Saad, M. Bennis, Y.-H. Nam, M. Debbah, A tutorial on UAVs for wireless networks: Applications, challenges, and open problems, *IEEE Commun. Surv. Tutor.* 21 (3) (2019) 2334–2360.
- [4] L. Gupta, R. Jain, G. Vaszkun, Survey of important issues in UAV communication networks, *IEEE Commun. Surv. Tutor.* 18 (2) (2016) 1123–1152.
- [5] B. Jang, M. Kim, G. Harerimana, J.W. Kim, Q-learning algorithms: A comprehensive classification and applications, *IEEE Access* 7 (2019) 133653–133667.
- [6] J. Liu, Q. Wang, C. He, K. Jaffrès-Runser, Y. Xu, Z. Li, Y. Xu, QMR:Q-learning based multi-objective optimization routing protocol for flying ad hoc networks, *Comput. Commun.* 150 (2020) 304–316.
- [7] C. He, Q. Wang, Y. Xu, J. Liu, Y. Xu, A Q-learning based cross-layer transmission protocol for MANETs, in: 2019 IEEE International Conferences on Ubiquitous Computing and Communications (IUCC) and Data Science and Computational Intelligence (DSCI) and Smart Computing, Networking and Services, SmartCNS, 2019, pp. 580–585.
- [8] M.Y. Arafat, S. Moh, A Q-learning-based topology-aware routing protocol for flying ad hoc networks, *IEEE Internet Things J.* 9 (3) (2022) 1985–2000.
- [9] L.A.L. da Costa, R. Kunst, E. Pignaton de Freitas, Q-FANET:Improved Q-learning based routing protocol for FANETs, *Comput. Netw.* 198 (2021) 108379.
- [10] W.-S. Jung, J. Yim, Y.-B. Ko, QGeo: Q-learning-based geographic ad hoc routing protocol for unmanned robotic networks, *IEEE Commun. Lett.* 21 (10) (2017) 2258–2261.
- [11] Y. Bai, H. Zhao, X. Zhang, Z. Chang, R. Jäntti, K. Yang, Toward autonomous multi-UAV wireless network: A survey of reinforcement learning-based approaches, *IEEE Commun. Surv. Tutor.* 25 (4) (2023) 3038–3067.
- [12] L.R. Faganello, R. Kunst, C.B. Both, L.Z. Granville, J. Rochol, Improving reinforcement learning algorithms for dynamic spectrum allocation in cognitive sensor networks, in: 2013 IEEE Wireless Communications and Networking Conference, WCNC, 2013, pp. 35–40.
- [13] Y. Cui, Q. Zhang, Z. Feng, Z. Wei, C. Shi, H. Yang, Topology-aware resilient routing protocol for FANETs: An adaptive Q-learning approach, *IEEE Internet Things J.* 9 (19) (2022) 18632–18649.
- [14] X. Qiu, Y. Xie, Y. Wang, L. Ye, Y. Yang, QLGR: A Q-learning-based geographic FANET routing algorithm based on multi-agent reinforcement learning, *KSII Trans. Internet Inf. Syst.* 15 (11) (2021) 4244–4274.
- [15] Y.N. Chen, N.Q. Lyu, G.H. Song, B.W. Yang, A traffic-aware Q-network enhanced routing protocol based on GPSR for unmanned aerial vehicle ad hoc networks, 2020, pp. 1308–1320.
- [16] J.W. Lim, Y.-B. Ko, Q-learning based stepwise routing protocol for multi-UAV networks, in: 2021 International Conference on Artificial Intelligence in Information and Communication, ICAIC, 2021, pp. 307–309.
- [17] M. Zhang, C. Dong, S. Feng, X. Guan, H. Chen, Q. Wu, Adaptive 3D routing protocol for flying ad hoc networks based on prediction-driven Q-learning, *China Commun.* 19 (5) (2022) 302–317.
- [18] N.P. Sharvari, D. Das, J. Bapat, D. Das, Improved Q-learning-based multi-hop routing for UAV-assisted communication, *IEEE Trans. Netw. Serv. Manag.* 22 (2) (2025) 1330–1344.
- [19] M.L. Puterman, Chapter 8 Markov decision processes, in: *Stochastic Models*, in: *Handbooks in Operations Research and Management Science*, vol. 2, Elsevier, 1990, pp. 331–434.
- [20] C. Qu, F.B. Sorbelli, R. Singh, P. Calyam, S.K. Das, Environmentally-aware and energy-efficient multi-drone coordination and networking for disaster response, *IEEE Trans. Netw. Serv. Manag.* 20 (2) (2023) 1093–1109.
- [21] M.M. Alam, S. Moh, Q-learning-based routing inspired by adaptive flocking control for collaborative unmanned aerial vehicle swarms, *Veh. Commun.* 40 (2023) 100572.
- [22] C. Zhang, V. Lesser, Coordinating multi-agent reinforcement learning with limited communication, in: *Proceedings of the 2013 International Conference on Autonomous Agents and Multi-Agent Systems*, 2013, pp. 1101–1108.
- [23] S.R. Chowdhury, A. Gopalan, O.-A. Maillard, Reinforcement learning in parametric mdps with exponential families, in: *International Conference on Artificial Intelligence and Statistics*, PMLR, 2021, pp. 1855–1863.
- [24] K. Zhang, Z. Yang, T. Başar, Multi-agent reinforcement learning: A selective overview of theories and algorithms, in: *Handbook of Reinforcement Learning and Control*, vol. 2, Springer, 2021, pp. 319–463, ser. 325.
- [25] S. Levine, V. Koltun, Proc. 30th Int. Conf. Mach. Learn. 28 (3) (2013) 1–9.
- [26] C. Guestrin, M. Lagoudakis, R. Parr, Coordinated reinforcement learning, in: *ICML*, vol. 2, Citeseer, 2002, pp. 227–234.
- [27] N.P. Sharvari, D. Das, J. Bapat, D. Das, Connectivity and collision constrained opportunistic routing for emergency communication using UAV, *Comput. Netw.* 220 (2023) 109468.
- [28] M.Y. Arafat, S. Moh, Location-aided delay tolerant routing protocol in UAV networks for post-disaster operation, *IEEE Access* 6 (2018) 59891–59906.
- [29] V.V. Chetlur, H.S. Dhillon, Downlink coverage analysis for a finite 3-D wireless network of unmanned aerial vehicles, *IEEE Trans. Commun.* 65 (10) (2017) 4543–4558.
- [30] I. Atzeni, J. Arnau, M. Kountouris, Downlink cellular network analysis with LOS/NLOS propagation and elevated base stations, *IEEE Trans. Wirel. Commun.* 17 (1) (2018) 142–156.
- [31] W. Liu, G. Niu, Q. Cao, M.-O. Pun, J. Chen, Particle swarm optimization for interference-limited unmanned aerial vehicle-assisted networks, *IEEE Access* 8 (2020) 174342–174352.

- [32] C. Guestrin, D. Koller, R. Parr, Adv. Neural Inf. Process. Syst 14.
- [33] W.-L. Yeow, C.-K. Tham, W.-C. Wong, Energy efficient multiple target tracking in wireless sensor networks, IEEE Trans. Veh. Technol. 56 (2) (2007) 918–928.
- [34] M. Haenggi, On distances in uniformly random networks, IEEE Trans. Inform. Theory 51 (10) (2005) 3584–3586.
- [35] S. Vural, E. Ekici, On multihop distances in wireless sensor networks with random node locations, IEEE Trans. Mob. Comput. 9 (4) (2010) 540–552.
- [36] I.S. Gradshteyn, I.M. Ryzhik, Academic Press, 2014.



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